

Task 1 – Classes: Data Validation

AI Prompt

Create a simple Python program for a Student class with name, roll_no, and marks. Use a constructor, validate marks

Python Code

```
class Student:
    def __init__(self, name, roll_no, marks):
        # Validate marks
        if marks < 0 or marks > 100:
            raise ValueError("Marks must be between 0 and 100")

        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    # Check whether the student passed
    def is_pass(self):
        return self.marks >= 40

student1 = Student("Sai", 101, 75)
student2 = Student("Ravi", 102, 35)

print("Student 1")
print("Name:", student1.name)
print("Roll No:", student1.roll_no)
print("Marks:", student1.marks)
print("Result:", "Pass" if student1.is_pass() else "Fail")

print()

print("Student 2")
print("Name:", student2.name)
print("Roll No:", student2.roll_no)
print("Marks:", student2.marks)
print("Result:", "Pass" if student2.is_pass() else "Fail")
```

VS Code PowerShell Output

```
Student 1
Name: Sai
Roll No: 101
Marks: 75
Result: Pass

Student 2
Name: Ravi
Roll No: 102
Marks: 35
Result: Fail
```

Code Explanation

The class stores student information, validates marks, and uses `is_pass()` to check the 40-mark passing condition.

Task 2 – Loops: Pattern Generation

AI Prompt

Generate a Python program that prints a right-angled triangle star pattern using a for loop and then the same pattern

Python Code

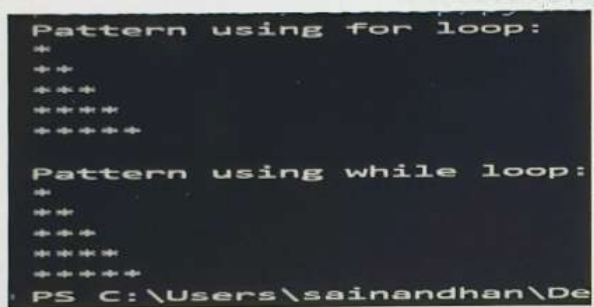
```
print("Pattern using for loop:")

for i in range(1, 6):
    print("*" * i)

print("\nPattern using while loop:")

i = 1
while i <= 5:
    print("*" * i)
    i += 1
```

VS Code PowerShell Output



```
Pattern using for loop:
*
**
***
****
*****

Pattern using while loop:
*
**
***
****
*****

PS C:\Users\sainandhan\De
```

Code Explanation

The for loop uses range(1, 6). The while loop starts at 1 and increments until 5. Both produce the same right-angled triangle.

Task 3 – Conditional Statements: Number Analysis

AI Prompt

Write a Python function `check_number()` that uses if-elif-else to classify a number as positive, negative, or zero. T

Python Code

```
def check_number(number):  
    if number > 0:  
        return "Positive"  
    elif number < 0:  
        return "Negative"  
    else:  
        return "Zero"  
  
numbers = [25, -10, 0, 7, -3]  
  
for number in numbers:  
    print(number, "is", check_number(number))
```

VS Code PowerShell Output

```
25 is Positive  
-10 is Negative  
0 is Zero  
7 is Positive  
-3 is Negative
```

Code Explanation

The if condition handles positive numbers, elif handles negative numbers, and else handles zero.

Task 4 – Nested Conditionals

AI Prompt

Create a Python function `check_discount(age, is_member)` using nested if statements. Age 60 or above qualifies for a

Python Code

```
def check_discount(age, is_member):  
    if age >= 60:  
        if is_member:  
            return "Senior discount + Additional member discount"  
        else:  
            return "Senior discount"  
    else:  
        if is_member:  
            return "Additional member discount"  
        else:  
            return "No discount"  
  
print("Age 65, Member:", check_discount(65, True))  
print("Age 65, Non-member:", check_discount(65, False))  
print("Age 30, Member:", check_discount(30, True))  
print("Age 30, Non-member:", check_discount(30, False))
```

VS Code PowerShell Output

```
C:\Users\sainandhan\Desktop\python practise> python 122.py  
Age 65, Member: Senior discount + Additional member discount  
Age 65, Non-member: Senior discount  
Age 30, Member: Additional member discount  
Age 30, Non-member: No discount  
PS C:\Users\sainandhan\Desktop\python practise>
```

Code Explanation

The outer condition checks age. The nested condition checks membership and selects the correct discount result.

Task 5 – Class: Mathematical Operations

AI Prompt

Create a Python Circle class that accepts a radius and provides methods for area and circumference. Use $\text{area} = \pi \times r^2$.

Python Code

```
import math

class Circle:
    def __init__(self, radius):
        # Validate radius
        if radius < 0:
            raise ValueError("Radius cannot be negative")
        self.radius = radius

    def area(self):
        return math.pi * self.radius ** 2

    def circumference(self):
        return 2 * math.pi * self.radius

circle = Circle(5)

print("Radius:", circle.radius)
print("Area:", round(circle.area(), 2))
print("Circumference:", round(circle.circumference(), 2))
```

VS Code PowerShell Output

```
Radius: 5
Area: 78.54
Circumference: 31.42
```

Code Explanation

The Circle class stores the radius. The area and circumference methods use the standard formulas, with results rounded to two decimal places.

Conclusion

All five assignment tasks are completed in order. Each task contains an AI prompt, separate Python code, VS Code PowerShell output using the requested Python executable path, and a brief explanation.